Year 13 Maths - Proof and mathematical communication

Similar questions - Proof by contradiction and by cases

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Prove by contradiction that there is no smallest positive rational number. **[3]**

(b) Prove by contradiction that $\sqrt{3}$ is irrational. **[5]**

2 Any integer can be written in the form $3k$, $3k+1$ or $3k+2$, where k is an integer.

(a) Prove that the square of any integer is of the form $3m$ or $3m+1$. **[4]**

(b) Hence prove that n^2+1 is never divisible by 3. **[2]**

(c) Prove that if a^2+b^2 is divisible by 3, then both a and b are divisible by 3. **[3]**

Question 1

(a) Suppose there is a smallest positive rational number q. Then $\frac{q}{2}$ is also a positive rational number and $\frac{q}{2} < q$, which contradicts the assumption. So there is no smallest positive rational

(b) Suppose $\sqrt{3} = \frac{a}{b}$ where a, b are integers with no common factor. Then $a^2 = 3b^2$, so a^2 is a multiple of 3, and therefore a is a multiple of 3 (a number not divisible by 3 has a square not divisible by 3 by Question 2(a)). Let $a = 3c$. Then $9c^2 = 3b^2$, so $b^2 = 3c^2$ and b is also a multiple of 3. This contradicts a and b having no common factor. So $\sqrt{3}$ is irrational

Question 2

(a) $(3k)^2 = 9k^2 = 3(3k^2)$. $(3k+1)^2 = 9k^2+6k+1 = 3(3k^2+2k)+1$. $(3k+2)^2 = 9k^2+12k+4 = 3(3k^2+4k+1)+1$. So squares are $3m$ or $3m+1$

(b) n^2+1 is $3m+1$ or $3m+2$, never a multiple of 3

(c) Squares have remainder 0 or 1 on division by 3. The sum of two squares has remainder 0 only when both remainders are 0 $(0+0)$; $0+1$, $1+0$ and $1+1=2$ are not multiples of 3. So both a^2 and b^2, and hence a and b, are multiples of 3